7C Volume, Capacity of Prisms and Time Zones

Standard Worksheet

Name: _____ Class: _____ Date: _____

PART A · FLUENCY & SKILL PRACTICE

Show complete step-by-step working in your workbook

Q1. Convert each volume or capacity measurement to the specified unit:

(a) $650\ \text{cm}^3$ to mL (b) $14\,500\ \text{cm}^3$ to litres (L) (c) $3.8\ \text{m}^3$ to litres (L) (d) $25\,000\ \text{L}$ to cubic metres (m^3)

Q2. Calculate the volume $V = Ah$ of each right prism:

(a) Rectangular prism: $12\ \text{cm} \times 8\ \text{cm} \times 5\ \text{cm}$ (b) Prism with cross-sectional area $A = 35\ \text{cm}^2$, length $h = 14\ \text{cm}$ (c) Triangular prism: triangle base $10\ \text{cm}$, height $7\ \text{cm}$, prism length $16\ \text{cm}$ (d) Trapezoidal prism: $a = 4\ \text{m}, b = 8\ \text{m}, h_t = 3\ \text{m}$, length $10\ \text{m}$

Q3. Convert between 12-hour and 24-hour time and calculate elapsed durations:

(a) Write $3{:}45\ \text{pm}$ in 24-hour time (b) Write $00{:}25$ in 12-hour am/pm time (c) Duration from $09{:}35$ to $14{:}15$ (d) Time $2\ \text{h}\ 50\ \text{min}$ after $11{:}25\ \text{am}$

Q4. When it is $2{:}00\ \text{pm}$ AEST (UTC$+10$) in Melbourne in July (standard winter time), find the local time in:

(a) Brisbane (AEST, UTC$+10$) (b) Adelaide and Darwin (ACST, UTC$+9.5$) (c) Perth (AWST, UTC$+8$) (d) Auckland, NZ (NZST, UTC$+12$)

PART B · PROBLEM-SOLVING & APPLICATIONS

Write clear mathematical statements with units

Q5. A rectangular garden Lap Pool in Echuca is $15\ \text{m}$ long, $3\ \text{m}$ wide, and has a uniform depth of $1.4\ \text{m}$. Calculate (a) its volume in m^3 and (b) its capacity in litres (L).

Q6. A wedge of Gippsland cheddar cheese is shaped like a right triangular prism with a right-angled triangle base of legs $9\ \text{cm}$ and $12\ \text{cm}$ and a thickness of $6\ \text{cm}$. Find the volume of the cheese wedge in cm^3.

Q7. During January (Daylight Saving), Sydney is on AEDT (UTC$+11$) and Brisbane is on AEST (UTC$+10$). A flight departs Sydney at $8{:}20\ \text{am}$ AEDT and takes $1\ \text{hour}\ 35\ \text{minutes}$ to reach Brisbane. What is the local arrival time in Brisbane (AEST)?

Q8. A rectangular rainwater tank has a base measuring $2.5\ \text{m}$ by $1.6\ \text{m}$. When $6000\ \text{L}$ ($6\ \text{m}^3$) of water is pumped into the empty tank, to what depth h (in metres) does the water rise?

PART C · REASONING & CHALLENGE

Justify your conclusions using algebraic or geometric reasoning

Q9. A solid metal cube of side length $10\ \text{cm}$ is submerged completely inside a rectangular tank of base area $250\ \text{cm}^2$ partially filled with water. By how many centimetres does the water level rise? Show your reasoning.

Q10. A live cricket test match in Perth starts at $10{:}30\ \text{am}$ AWST (UTC$+8$). Explain why viewers in Melbourne watching in December (AEDT, UTC$+11$) see the first ball bowled at $1{:}30\ \text{pm}$, whereas if it were played in winter (AEST, UTC$+10$) they would see it at $12{:}30\ \text{pm}$.